Homework 2

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The research problem is that, despite extensive literature on labor mobility, little is known about inter-industry labor flow networks, leaving human capital specificity and skill transferability unclear. Previous measures (co-location, co-exports, input-output) captured indirect economic overlap rather than direct similarity in skill requirements. The paper asks what cross-industry labor flows reveal about human capital specificity and whether they measure skill-relatedness reliably. Its contribution is seven stylized facts from German data, showing that labor flows are concentrated, stable, and distinct from geographic co-location, and that labor-based skill-relatedness better predicts regional industry diversification than traditional metrics.

Hane-Weijman et al. (2021) extend Neffke et al. (2017) by examining occupations rather than industries in Sweden and investigating how relatedness and complexity changes relate to employment growth. They find that occupational relatedness positively predicts employment growth in Swedish local labor markets, while complexity changes have no overall effect, although entering more complex occupations becomes more favorable after the crisis.

Buyukyazici et al. (2023) extend Neffke et al. (2017) by combining labor-flow-based relatedness with direct workplace skill data, distinguishing skill types and incorporating complexity into regional industrial diversification. They find that higher skill-relatedness and complexity increase the likelihood of gaining a new comparative advantage and reduce the likelihood of losing one, depending on skill type and industry.

Previous work shows that workers mainly switch between industries with similar skill requirements and that immigrants from countries that export a product increase the host country's likelihood of exporting it. However, we do not know whether immigrants' knowledge utilization depends on the similarity between their skills and the host region's existing industries. This matters because regions and policymakers need to decide whether to attract migrants whose skills match local labor demand or complement it.

Research question: Does the skill similarity between immigrants' origin industries/occupations and a region's existing industries predict whether the region develops a new industry in the coming years?

This question could be answered using individual microdata from national statistical offices, including workers' occupations, workplaces, and industries. Policymakers and multinational corporations could use the findings.